

## 37.5 Zero inflation

Many count data sets have far more zeros than a Poisson or negative binomial model with the right mean can produce. The usual explanation is that zeros come from two places: some units are not at risk at all — they do not smoke, do not drive, are not eligible — and contribute a zero with certainty, while the rest contribute one only when the count happens to be zero. This section writes that story down as a model, does the same for its rival, and asks how the two can be told apart.

### 37.5.1 Two ways to make extra zeros

#### Definition 37.5.1 — Zero-inflated and hurdle models

Let  $f(y; \lambda)$  be a count distribution with parameter  $\lambda$ , such as the Poisson or the NB2 family of Definition 37.4.1, and let  $\mathbf{z}_{(i)}$  and  $\mathbf{x}_{(i)}$  be rows of two model matrices, which may be equal, overlapping or disjoint.

- (a) The **zero-inflated** model gives  $Y_i$  the mass function

$$\Pr(Y_i = y) = \begin{cases} \pi_i + (1 - \pi_i)f(0; \lambda_i), & y = 0, \\ (1 - \pi_i)f(y; \lambda_i), & y = 1, 2, \dots, \end{cases} \quad (37.5.1)$$

with  $\text{logit } \pi_i = \mathbf{z}_{(i)}^\top \boldsymbol{\gamma}$  and  $\log \lambda_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . Equivalently, a latent indicator  $Z_i$  is one with probability  $\pi_i$ , and  $Y_i = 0$  when  $Z_i = 1$  while  $Y_i \sim f(\cdot; \lambda_i)$  when  $Z_i = 0$ . Taking  $f$  Poisson gives the **zero-inflated Poisson** (ZIP), and  $f$  negative binomial the **zero-inflated negative binomial** (ZINB).

- (b) The **hurdle** model gives  $Y_i$  the mass function

$$\Pr(Y_i = y) = \begin{cases} 1 - \rho_i, & y = 0, \\ \rho_i \frac{f(y; \lambda_i)}{1 - f(0; \lambda_i)}, & y = 1, 2, \dots, \end{cases} \quad (37.5.2)$$

with  $\text{logit } \rho_i = \mathbf{z}_{(i)}^\top \boldsymbol{\gamma}$  and  $\log \lambda_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ : a binary model decides whether the count is positive, and a **zero-truncated** version of  $f$  decides how large it is.

These are different stories. The zero-inflated model says there are two kinds of unit and a zero does not reveal which; the hurdle model says there is one kind of unit and two decisions, whether to cross the threshold and how far to go once across. Which is right is a subject-matter question, and the data distinguish them only indirectly.

#### Proposition 37.5.1 — Properties of the two models

In the notation of Definition 37.5.1, write  $\mu_i = E(Y_i)$  and  $\mathbf{1}_i = \mathbf{1}\{y_i > 0\}$ . The zero-

inflated log-likelihood is

$$\begin{aligned} & \sum_{i:y_i=0} \log\{\pi_i + (1 - \pi_i)f(0; \lambda_i)\} \\ & + \sum_{i:y_i>0} [\log(1 - \pi_i) + \log f(y_i; \lambda_i)], \end{aligned}$$

and the hurdle one is

$$\begin{aligned} & \sum_i [\mathbf{1}_i \log \rho_i + (1 - \mathbf{1}_i) \log(1 - \rho_i)] \\ & + \sum_{i:y_i>0} [\log f(y_i; \lambda_i) - \log\{1 - f(0; \lambda_i)\}]. \end{aligned}$$

Then:

- (a) for the zero-inflated model with  $f$  Poisson,

$$\mu_i = (1 - \pi_i)\lambda_i, \quad \text{Var}(Y_i) = \mu_i(1 + \pi_i\lambda_i), \quad (37.5.3)$$

so a zero-inflated Poisson is always overdispersed relative to its own mean unless  $\pi_i = 0$ ;

- (b) the hurdle log-likelihood separates,  $\ell(\boldsymbol{\gamma}, \boldsymbol{\beta}, \cdot) = \ell_{\text{binary}}(\boldsymbol{\gamma}) + \ell_{\text{positive}}(\boldsymbol{\beta}, \cdot)$ , so the two parts can be fitted independently, and their estimates are asymptotically independent; the zero-inflated log-likelihood does not separate;
- (c) the hurdle model can represent **zero deflation**,  $\rho_i > 1 - f(0; \lambda_i)$ , while [Eq. 37.5.1](#) needs  $\pi_i \geq 0$  and can only add zeros; and, for the hurdle model, if  $\mathbf{z}_{(i)}$  contains an intercept then its fitted probabilities of a zero average to the observed proportion of zeros, whatever else is in the binary part. The zero-inflated model satisfies no such identity;
- (d) the zero-inflated model contains  $f$  itself as the limiting case  $\pi_i \equiv 0$ , which lies outside the range where the usual expansion is valid, so a likelihood ratio test of  $f$  against its zero-inflated version is *not* asymptotically  $\chi^2$ ; nor is the Vuong statistic of [Eq. 37.5.5](#) valid for such a comparison.

*Proof.*

- (a) With the latent representation,  $E(Y_i) = E\{E(Y_i | Z_i)\} = (1 - \pi_i)\lambda_i = \mu_i$ . Since  $E(Y_i | Z_i) = \lambda_i(1 - Z_i)$  has variance  $\lambda_i^2\pi_i(1 - \pi_i)$ , the law of total variance gives

$$\begin{aligned} \text{Var}(Y_i) &= E\{\text{Var}(Y_i | Z_i)\} + \text{Var}\{E(Y_i | Z_i)\} \\ &= (1 - \pi_i)\lambda_i + \lambda_i^2\pi_i(1 - \pi_i). \end{aligned}$$

Factoring out  $(1 - \pi_i)\lambda_i = \mu_i$  gives  $\text{Var}(Y_i) = \mu_i(1 + \pi_i\lambda_i)$ , which exceeds  $\mu_i$  whenever  $\pi_i > 0$ .

(b) With  $\mathbf{1}_i = \mathbf{1}\{y_i > 0\}$ , Eq. 37.5.2 gives

$$\begin{aligned} \ell = \sum_i & \left[ \mathbf{1}_i \log \rho_i + (1 - \mathbf{1}_i) \log(1 - \rho_i) \right] \\ & + \sum_{i: y_i > 0} \left[ \log f(y_i; \lambda_i) - \log\{1 - f(0; \lambda_i)\} \right]. \end{aligned}$$

The first sum involves only  $\boldsymbol{\gamma}$  and the second only  $\boldsymbol{\beta}$  and any shape parameter of  $f$ , with unconstrained parameter spaces, so the joint maximum is the pair of separate maxima and the information is block diagonal. For Eq. 37.5.1 the contribution of a zero is  $\log\{\pi_i + (1 - \pi_i)f(0; \lambda_i)\}$ , which involves both parameter vectors inside one logarithm and does not separate.

- (c) In Eq. 37.5.2,  $\rho_i$  is a free probability, so  $\Pr(Y_i = 0) = 1 - \rho_i$  can take any value in  $(0, 1)$ ; in Eq. 37.5.1,  $\Pr(Y_i = 0) = \pi_i + (1 - \pi_i)f(0; \lambda_i) \geq f(0; \lambda_i)$  for  $\pi_i \in [0, 1]$ . The first sum in the display in (b) is the log-likelihood of a logistic regression of  $\mathbf{1}_i$  on  $\mathbf{z}_{(i)}$ , whose likelihood equations are  $\sum_i (\mathbf{1}_i - \hat{\rho}_i) \mathbf{z}_{(i)} = \mathbf{0}$  by Theorem 34.3.1 with the canonical link. If  $\mathbf{z}_{(i)}$  contains an intercept, one of those equations reads  $\sum_i \hat{\rho}_i = \sum_i \mathbf{1}_i$ , so  $n^{-1} \sum_i (1 - \hat{\rho}_i)$  is the observed proportion of zeros. The zero-inflated likelihood equation for the inflation intercept is instead  $\sum_i \hat{\pi}_i = \sum_i \hat{w}_i$ , with  $\hat{w}_i$  the posterior weights Eq. 37.5.4, which says nothing about  $\sum_i \widehat{\Pr}(Y_i = 0)$ .
- (d) Setting  $\pi_i \equiv 0$  in Eq. 37.5.1 recovers  $f$ , but how the  $\chi^2$  argument fails depends on the parameterization. If inflation is a single probability  $\pi$ , the null sits on the boundary of  $[0, 1]$  and Proposition 37.4.2(e) applies verbatim. If  $\text{logit } \pi_i = \mathbf{z}_{(i)}^\top \boldsymbol{\gamma}$ , the null is reached only as the intercept tends to  $-\infty$ , and under it the remaining coefficients of  $\boldsymbol{\gamma}$  are unidentified — a harder non-regularity, where no fixed mixture of  $\chi^2$  laws is the limit. Either way the calibration fails. The Vuong statistic assumes strictly non-nested models, so that the mean of its summands is nonzero under the null; when they are nested that mean is zero and the limit is not normal.

■

Part (a) matters when choosing among the four combinations. A zero-inflated Poisson is already overdispersed, so excess zeros with no other anomaly can be described either by a ZIP or by a negative binomial; what a ZIP cannot do is produce a long right tail without also producing extra zeros. Zero inflation on a negative binomial addresses zeros and tail separately, which is why ZINB is the workhorse.

## 37.5.2 Fitting

The hurdle model needs no new machinery: fit a logistic regression to  $\mathbf{1}\{y_i > 0\}$ , then maximize  $\sum_{y_i > 0} [\log f(y_i; \lambda_i) - \log\{1 - f(0; \lambda_i)\}]$  numerically over the positive observations. The zero-inflated model does not separate, but its latent representation invites the EM algorithm, which is what Lambert used. The E step computes the posterior probability that observation  $i$  came from the degenerate component,

$$w_i = \Pr(Z_i = 1 \mid Y_i = y_i) = \begin{cases} \frac{\pi_i}{\pi_i + (1 - \pi_i)f(0; \lambda_i)}, & y_i = 0, \\ 0, & y_i > 0, \end{cases} \quad (37.5.4)$$

by Bayes's rule. The M step maximizes the expected complete-data log-likelihood, which *does* separate: a logistic regression of the fractional response  $w_i$  on  $\mathbf{Z}$ , and a count regression of  $y_i$  on  $\mathbf{X}$  with prior weights  $1 - w_i$ . Both are weighted fits already in hand. The M step need only *increase* that expected log-likelihood, not maximize it, so a single weighted least squares step for  $\beta$  and a single Newton step for the shape suffice; either way the observed-data log-likelihood increases at every iteration ([Exercise \(B4\)](#)).

### 37.5.3 An example

#### Example. (Five models for the physician visits)

A proportion 0.3124 of the 20190 person-years record no physician visit; the Poisson model of Example (Physician visits in a health insurance experiment) predicts 0.0809.

**Table 37.5.1.** Five count models for the physician-visit data, all with the same eight regressors; the zero-inflated models use them in both parts. The observed proportion of zeros is 0.3124.

| model     | parameters | log-likelihood | AIC      | BIC      | fitted<br>$\Pr(Y = 0)$ |
|-----------|------------|----------------|----------|----------|------------------------|
| Poisson   | 8          | -62737.3       | 125490.7 | 125554.0 | 0.0809                 |
| NB2       | 9          | -43468.5       | 86955.0  | 87026.3  | 0.3200                 |
| ZIP       | 16         | -54978.6       | 109989.1 | 110115.8 | 0.3125                 |
| ZINB      | 17         | -43391.8       | 86817.5  | 86952.0  | 0.3255                 |
| hurdle NB | 17         | -43244.6       | 86523.2  | 86657.7  | 0.3124                 |

Three things stand out. First, zero inflation alone does not save the Poisson: the ZIP reproduces the zeros almost exactly, at 0.3125, and still trails the plain negative binomial by more than eleven thousand in log-likelihood, because the trouble here is the whole shape of the distribution and not only its zeros. Second, the negative binomial *already* matches the zeros, at 0.3200 against 0.3124, as [Exercise \(B2\)](#) predicts: heterogeneity in the rate produces excess zeros as a by-product, and adding zero inflation gains 153.5 in twice the log-likelihood for eight parameters while slightly *overshooting*, at 0.3255. Third, the hurdle model wins on both AIC and BIC, and by construction reproduces the zero proportion exactly.

The hurdle model wins because of the positive counts, not the zeros. Splitting each log-likelihood at  $y = 0$  shows it: on the zeros the hurdle model scores -6966.1 against the ZINB's -6757.5, a loss of 208.6; on the positive counts -36278.5 against -36634.2, a gain of 355.7, for a net 147.2. Above two visits the two are indistinguishable in [Figure 37.5.1](#); the difference is in how mass is arranged over the small positive counts, and the hurdle model buys that freedom by giving the truncated component its own shape,  $\hat{\kappa} = 0.557$  against 0.838 in the ZINB. Estimated zero inflation in the ZINB averages only 0.0475: it too spends its extra parameters re-shaping the low counts, which is what a hurdle model does directly.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats
```

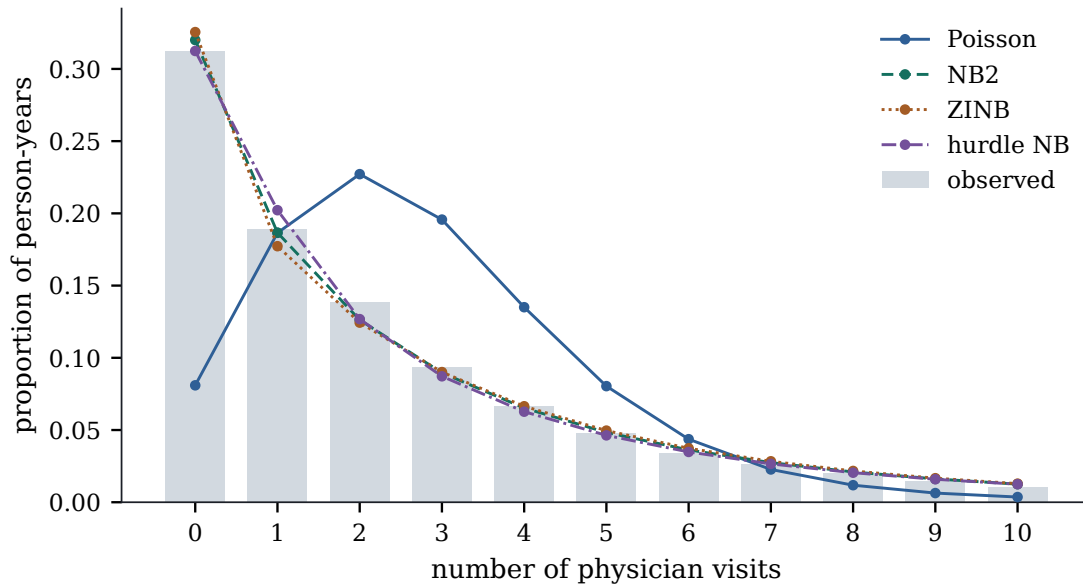


Figure 37.5.1: Observed distribution of physician visits (bars) and the fitted marginal distributions  $n^{-1} \sum_i \Pr(Y_i = k)$  under four models. The Poisson is hopeless at zero and too heavy in the middle; the other three are close to each other and to the data.

```
data = sm.datasets.randhie.load_pandas().data
y = data["mdvis"].to_numpy(float)
names = ["lncoins", "idp", "physlm", "disea", "hlthg", "hlthf", "hlthp"]
X = np.column_stack([np.ones(len(y))] + [data[v].to_numpy(float) for v in names])
n, p = X.shape

def poisson_irls(X, y, w=None, start=None, tol=1e-11, maxit=200):
    """Weighted Poisson regression with a log link; w are prior weights."""
    w = np.ones(len(y)) if w is None else w
    beta = np.zeros(X.shape[1]) if start is None else start.copy()
    if start is None:
        beta[0] = np.log(max((w * y).sum() / w.sum(), 1e-8))
    for _ in range(maxit):
        mu = np.exp(X @ beta)
        XW = X * (w * mu)[:, None]
        step = np.linalg.solve(X.T @ XW, XW.T @ (X @ beta + (y - mu) / mu))
        if np.max(np.abs(step - beta)) < tol:
            return step
        beta = step
    return beta

def logistic_irls(X, r, tol=1e-11, maxit=200):
    """Logistic regression with a response r in [0, 1] (a fractional response is
    allowed)."""
    gamma = np.zeros(X.shape[1])
    for _ in range(maxit):
        pi = 1 / (1 + np.exp(-X @ gamma))
        v = np.clip(pi * (1 - pi), 1e-10, None)
        XW = X * v[:, None]
        step = np.linalg.solve(X.T @ XW, XW.T @ (X @ gamma + (r - pi) / v))
        if np.max(np.abs(step - gamma)) < tol:
            return step
```

```

        gamma = step
    return gamma

def nb_logpmf(y, mu, kappa):
    """log P(Y = y) for the negative binomial with mean mu and shape kappa."""
    return (special.gammaln(y + kappa) - special.gammaln(kappa) - special.gammaln(y
+ 1)
            + kappa * np.log(kappa / (kappa + mu)) + y * np.log(mu / (mu + kappa)))

def poisson_logpmf(y, mu):
    return y * np.log(mu) - mu - special.gammaln(y + 1)

def nb_step(X, y, beta, kappa, w):
    """One IRLS step in beta and one Newton step in log kappa, at prior weights w.
    """
    mu = np.exp(X @ beta)
    XW = X * (w * mu / (1 + mu / kappa))[:, None] # NB2 working weights, log
    link
    beta = np.linalg.solve(X.T @ XW, XW.T @ (X @ beta + (y - mu) / mu))
    mu = np.exp(X @ beta)
    s = np.sum(w * (special.digamma(y + kappa) - special.digamma(kappa)
                    + np.log(kappa / (kappa + mu)) + 1 - (kappa + y) / (kappa + mu))
    )
    d = np.sum(w * (special.polygamma(1, y + kappa) - special.polygamma(1, kappa)
                    + 1 / kappa - 1 / (kappa + mu) - (mu - y) / (kappa + mu) ** 2))
    return beta, float(np.clip(np.exp(np.log(kappa) - s / (kappa * d)), 1e-4, 1e4))

def nb_fit(X, y, w=None, start=None, tol=1e-10, maxit=500):
    """Weighted NB2 maximum likelihood: iterate nb_step to convergence."""
    w = np.ones(len(y)) if w is None else w
    beta, kappa = (poisson_irls(X, y, w), 1.0) if start is None else start
    for _ in range(maxit):
        new_beta, new_kappa = nb_step(X, y, beta, kappa, w)
        done = max(np.abs(new_beta - beta).max(), abs(new_kappa - kappa)) < tol
        beta, kappa = new_beta, new_kappa
        if done:
            break
    return beta, kappa

beta_pois = poisson_irls(X, y)
loglik_pois = poisson_logpmf(y, np.exp(X @ beta_pois)).sum()
beta_nb, kappa_nb = nb_fit(X, y)
loglik_nb = nb_logpmf(y, np.exp(X @ beta_nb), kappa_nb).sum()
print(f"Poisson {loglik_pois:.1f}    NB2 {loglik_nb:.1f} (kappa {kappa_nb:.3f})")

```

[Run this code online](#)

```

def zero_inflated_em(X, Z, y, negbin=False, tol=1e-10, maxit=2000):
    """EM for a zero-inflated Poisson or NB2 model; returns (gamma, beta, kappa,
    loglik).

    gamma indexes the logistic model for the degenerate component, beta the log-
    linear

```

*model for the count component. The latent indicator is  $z_i = 1$  if observation  $i$  comes from the degenerate component at zero.*

```

"""
zero = y == 0
gamma = np.zeros(Z.shape[1])
beta, kappa = (poisson_irls(X, y), np.inf)
if negbin:
    beta, kappa = nb_fit(X, y)
previous = -np.inf
for _ in range(maxit):
    mu = np.exp(X @ beta)
    pi = 1 / (1 + np.exp(-Z @ gamma))
    log_count = nb_logpmf(y, mu, kappa) if negbin else poisson_logpmf(y, mu)
    # E step: the posterior probability that a zero came from the degenerate
    part
    w = np.where(zero, pi / (pi + (1 - pi) * np.exp(log_count)), 0.0)
    # M step: a fractional logistic regression and a weighted count regression
    gamma = logistic_irls(Z, w)
    if negbin:
        beta, kappa = nb_step(X, y, beta, kappa, 1 - w) # one step is enough
    else:
        beta = poisson_irls(X, y, 1 - w, start=beta)
    pi = 1 / (1 + np.exp(-Z @ gamma))
    mu = np.exp(X @ beta)
    log_count = nb_logpmf(y, mu, kappa) if negbin else poisson_logpmf(y, mu)
    loglik = np.sum(np.where(zero,
                             np.log(pi + (1 - pi) * np.exp(log_count)),
                             np.log1p(-pi) + log_count))
    if loglik - previous < tol * abs(loglik):
        break
    previous = loglik
return gamma, beta, kappa, loglik

gamma_zip, beta_zip, _, loglik_zip = zero_inflated_em(X, X, y)
gamma_zinb, beta_zinb, kappa_zinb, loglik_zinb = zero_inflated_em(X, X, y, negbin=
    True)
print(f"ZIP log-likelihood {loglik_zip:.2f}")
print(f"ZINB log-likelihood {loglik_zinb:.2f} kappa {kappa_zinb:.3f}")

```

[Run this code online](#)

```

def truncated_nb_loglik(theta, X, y):
    """Log-likelihood of the zero-truncated NB2 model, parameters (beta, log kappa).
    """
    mu = np.exp(np.clip(X @ theta[:-1], -20.0, 20.0))
    k = np.exp(theta[-1])
    log_p0 = k * np.log(k / (k + mu)) # log P(Y = 0) before
    truncation
    return np.sum(nb_logpmf(y, mu, k) - np.log(-np.expm1(log_p0)))

def hurdle_nb(X, y):
    """Logistic model for  $y > 0$ , zero-truncated NB2 for the positive counts."""
    gamma = logistic_irls(X, (y > 0).astype(float))
    pi = 1 / (1 + np.exp(-X @ gamma))
    loglik_zero = np.sum(np.where(y > 0, np.log(pi), np.log1p(-pi)))

```

```

Xp, yp = X[y > 0], y[y > 0]
beta0, kappa0 = nb_fit(Xp, yp)                                # an untruncated fit as the
start
out = optimize.minimize(lambda t: -truncated_nb_loglik(t, Xp, yp),
                        np.append(beta0, np.log(kappa0)), method="BFGS",
                        options={"gtol": 1e-5, "maxiter": 800})
return gamma, out.x[:-1], np.exp(out.x[-1]), loglik_zero - out.fun

gamma_h, beta_h, kappa_h, loglik_hurdle = hurdle_nb(X, y)
print(f"hurdle log-likelihood {loglik_hurdle:.2f}    kappa {kappa_h:.3f}")

```

[Run this code online](#)

### 37.5.4 Telling them apart

Table 37.5.1 compares fits, not mechanisms, and settles no question about which story is right. It is worth being clear about what the data can and cannot decide.

**Identification.** When  $\mathbf{z}_{(i)}$  and  $\mathbf{x}_{(i)}$  are the same, the two parts compete to explain the same zeros and the likelihood is often very flat: a larger  $\pi_i$  with a larger  $\lambda_i$  fits almost as well as a smaller  $\pi_i$  with a smaller one. The parameters are identified in principle, since the positive counts pin down  $\lambda_i$ , but standard errors can be large, the two parts' coefficients strongly correlated, and the fit sensitive to the start. Restricting  $\mathbf{z}_{(i)}$  to a few variables, or to an intercept, helps and is easier to defend.

**The Vuong test.** For two non-nested models fitted to the same data, Vuong (1989) proposed comparing the pointwise log-likelihood ratios  $m_i = \log f_1(y_i; \hat{\theta}_1) - \log f_2(y_i; \hat{\theta}_2)$  with

$$V = \frac{\sqrt{n} \bar{m}}{s_m}, \quad s_m^2 = \frac{1}{n-1} \sum_{i=1}^n (m_i - \bar{m})^2, \quad (37.5.5)$$

which is asymptotically standard normal when the two models are equally close to the truth, so that large  $|V|$  favours one or the other by its sign. For the physician visits, ZINB against the hurdle model gives  $V = -7.23$ : taken at face value it prefers the hurdle model, agreeing with AIC.

The test is often misused, routinely applied to compare a count model with its own zero-inflated version — exactly the nested case the theory excludes. Under the null  $\bar{m}$  and  $s_m$  both tend to zero, the ratio has no normal limit, and the reported  $p$ -value is meaningless. In [Example \(Five models for the physician visits\)](#) that comparison gives  $V = 5.90$ , a number with no calibration; Wilson (2015) documents the practice. A further difficulty, quantified by Schennach and Wilhelm (2017), is poor size even in the non-nested case when the two models are close.

```

mu_p, mu_nb = np.exp(X @ beta_pois), np.exp(X @ beta_nb)
pi_zinb = 1 / (1 + np.exp(-X @ gamma_zinb))
mu_zinb = np.exp(X @ beta_zinb)
pi_h = 1 / (1 + np.exp(-X @ gamma_h))
mu_h = np.exp(X @ beta_h)
p0_h = np.exp(kappa_h * np.log(kappa_h / (kappa_h + mu_h)))

def vuong(log_f1, log_f2):

```

```

    """Vuong's statistic for model 1 against model 2 (not valid for nested models).
    """
    m = log_f1 - log_f2
    return np.sqrt(len(m)) * m.mean() / m.std(ddof=1)

log_zinb = np.where(y == 0, np.log(pi_zinb + (1 - pi_zinb) * np.exp(nb_logpmf(0.0,
    mu_zinb, kappa_zinb))),
    np.log1p(-pi_zinb) + nb_logpmf(y, mu_zinb, kappa_zinb))
log_hurdle = np.where(y == 0, np.log1p(-pi_h),
    np.log(pi_h) + nb_logpmf(y, mu_h, kappa_h) - np.log1p(-p0_h))
log_nb = nb_logpmf(y, mu_nb, kappa_nb)

print(f"ZINB against NB2 (nested, so not valid):  V = {vuong(log_zinb, log_nb):6.2f}"
    ")
print(f"ZINB against the hurdle model:          V = {vuong(log_zinb, log_hurdle)
    :6.2f}")

```

[Run this code online](#)

**What does distinguish them.** By [Proposition 37.5.1\(c\)](#) the two differ in what they can say about the positive counts and about zero deflation, and those are where to look: comparing fitted and observed frequencies at  $y = 1, 2, 3$ , as in [Figure 37.5.1](#), is more informative than any single statistic.

*Warning.* Excess zeros are a symptom, not a diagnosis. Before reaching for a two-part model, check that the mean model is right, that the overdispersion is not heterogeneity a negative binomial would absorb, and that the zeros are not an artefact of the recording process: a question never asked, a period of non-observation, a value truncated on entry. The physician-visit data make the point. Of the two models, the hurdle likelihood factorizes and the zero-inflated one does not, which makes the hurdle easier to fit and to interpret; the zero-inflated model has the better story when there really are two kinds of unit.

## 37.5.5 Exercises

### 37.5.5.1 A. Check your understanding

#### Exercise A1.

A ZIP model has  $\pi = 0.3$  and  $\lambda = 4$ . Compute  $EY$ ,  $\text{Var } Y$ ,  $\Pr(Y = 0)$  and the ratio of variance to mean, and compare with a Poisson having the same mean.

### 37.5.5.2 B. Practice

#### Exercise B1.

Let  $Y$  be Poisson with mean  $\lambda$ , truncated to  $\{1, 2, \dots\}$ . Show that  $EY = \lambda/(1 - e^{-\lambda})$  and  $\text{Var } Y = EY\{1 + \lambda - EY\}$ , and deduce that the truncated distribution is underdispersed relative to its own mean.

*Solution.* Write  $c = 1 - e^{-\lambda}$ . The  $y = 0$  term contributes nothing to either moment, so  $EY = \lambda/c$  and  $E\{Y(Y-1)\} = \lambda^2/c$ , giving  $\text{Var } Y = \lambda^2/c + \lambda/c - \lambda^2/c^2 = EY(1 + \lambda - EY)$ . Since  $EY = \lambda/c > \lambda$ , the bracket is less than one.

### Exercise B2.

Derive the mean and variance of a ZINB with inflation probability  $\pi$ , count mean  $\lambda$  and shape  $\kappa$ . Show that for a given mean and variance there is a curve of  $(\pi, \lambda, \kappa)$  values giving the same first two moments, and explain what this implies for the precision of the estimates.

*Solution.*  $EY = (1 - \pi)\lambda$  and, by the law of total variance,  $\text{Var } Y = (1 - \pi)(\lambda + \lambda^2/\kappa) + \pi(1 - \pi)\lambda^2 = EY\{1 + \lambda(\pi + 1/\kappa)\}$ . The mean fixes  $(1 - \pi)\lambda$  and the variance  $\lambda(\pi + 1/\kappa)$ : two equations in three unknowns, so a one-dimensional family shares the first two moments. Only higher moments separate them, so the likelihood is flat along it and the estimates are strongly correlated with large standard errors.

### Exercise B3.

Fit a ZINB to the physician-visit data using only an intercept in the inflation part, and compare its AIC with the model of [Example \(Five models for the physician visits\)](#) that uses all the regressors in both parts. Which does AIC prefer, and what does the comparison say about the eight inflation coefficients?

### Exercise B4.

Show that the EM step described around [Eq. 37.5.4](#) increases the observed-data log-likelihood at every iteration, by writing the observed-data log-likelihood as the expected complete-data log-likelihood minus an entropy term and using Jensen's inequality.

## 37.5.5.3 C. Going deeper

### Exercise C1.

Take  $f$  Poisson,  $\mathbf{z}_{(i)} = \mathbf{x}_{(i)}$ , and a hurdle model whose binary part uses the complementary log-log link,  $\log\{-\log(1 - \rho_i)\} = \mathbf{z}_{(i)}^\top \boldsymbol{\gamma}$ , in place of the logit of [Definition 37.5.1\(b\)](#). Show that this model contains the plain Poisson as the submodel  $\boldsymbol{\gamma} = \boldsymbol{\beta}$ , so that a likelihood ratio test of the Poisson against it is a standard interior test on  $p$  restrictions, unlike the zero-inflated comparison of [Proposition 37.5.1\(d\)](#). Then show that the same is not true if the binary part uses the logit.

*Solution.* The hurdle model reduces to  $f$  when  $\rho_i = 1 - f(0; \lambda_i)$  for every  $i$ . For the Poisson that is  $1 - e^{-\lambda_i}$ , so  $-\log(1 - \rho_i) = \lambda_i$  and the complementary log-log link makes the constraint read  $\mathbf{z}_{(i)}^\top \boldsymbol{\gamma} = \log \lambda_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , that is  $\boldsymbol{\gamma} = \boldsymbol{\beta}$ : a linear restriction of  $p$  dimensions in the interior of  $\mathbb{R}^{2p}$ , so the usual  $\chi^2(p)$  limit applies. With the logit the constraint is  $\mathbf{x}_{(i)}^\top \boldsymbol{\gamma} = \log(e^{\lambda_i} - 1)$ , and since  $t \mapsto \log(e^t - 1)$  is not affine, the vector on the right lies in  $\mathcal{C}(\mathbf{X})$  only in degenerate cases such as  $\mathbf{X}\boldsymbol{\beta}$  constant. The Poisson is then not a smooth submodel at all and no likelihood ratio test of one against the other

exists: a model “containing” another is not the same as nesting it.

### Exercise C2.

In a zero-inflated model  $\beta_j$  is a log rate ratio for the count component, not for the response. Derive  $\partial \log E(Y_i)/\partial x_{ij}$  when  $x_j$  appears in both parts, and explain why reporting  $e^{\hat{\beta}_j}$  as “the effect on the expected count” is wrong.

*Solution.*  $E(Y_i) = (1 - \pi_i)\lambda_i$ , so  $\log E(Y_i) = \log(1 - \pi_i) + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  and

$$\frac{\partial \log E(Y_i)}{\partial x_{ij}} = \beta_j - \frac{1}{1 - \pi_i} \frac{\partial \pi_i}{\partial x_{ij}} = \beta_j - \pi_i \gamma_j,$$

using  $\partial \pi_i / \partial x_{ij} = \pi_i(1 - \pi_i)\gamma_j$  for the logit link. The effect on the expected response therefore mixes the two parts and depends on  $\pi_i$ , hence on all the covariates;  $e^{\hat{\beta}_j}$  is the rate ratio only within the non-degenerate component, a conditional statement about a latent class nobody observes.